

Module 3: Input and Output

3.1 Output with cout

- cout stands for character output and is used to display information on the screen
- It is defined in the iostream library
- The insertion operator << sends data to cout
Reading it aloud: "cout puts out the value on the right"
- Printing a string literal
cout << "Hello, World!";
- Printing a variable value
int age = 20;
cout << age;
- Printing a combination of text and variables
cout << "Age is: " << age;
- Moving to the next line
 - endl: flushes the output buffer and moves to a new line
 - \n inside a string: moves to a new line without flushing the buffer
 - \n is slightly faster than endl because it does not force a buffer flush
- Printing multiple values in one statement
 - cout << "Name: " << name << ", Age: " << age << endl;
- Common escape sequences used inside string literals
 - \n: newline
 - \t: horizontal tab
 - \: backslash character
 - ": double quote inside a string
 - 'single quote inside a character

3.2 Input with cin

- cin stands for character input and reads data typed by the user from the keyboard
- The extraction operator >> reads data from cin into a variable
- Reading it aloud: "cin extracts the value into the variable on the right"
- Reading an integer
int age;
cin >> age;
- Reading a floating point number
double salary;
cin >> salary;
- Reading a character
char grade;
cin >> grade;
- Reading multiple values in one statement
cin >> x >> y;

- The user must separate values with spaces or newlines
- What happens with wrong input type
- If the user types a letter when an int is expected, cin fails silently
- The variable is left unchanged or set to zero
- Input validation should always be done in real programs
- cin skips leading whitespace automatically before reading

3.3 Reading Strings with cin and getline

- The problem with cin for strings
- cin >> word reads only up to the first space
- If the user types "John Doe", cin reads only "John"
- Using getline() to read a full line including spaces
- Syntax: getline(cin, variableName);
- Must use a string variable, not a char array
- The buffer problem when mixing cin and getline
- When cin reads a number, the newline character (Enter key) remains in the buffer
- getline() reads that leftover newline and immediately returns an empty string
- Fix: use cin.ignore() after cin to discard the leftover newline before calling getline
- Example: cin >> age; cin.ignore(); getline(cin, name);

3.4 Formatted Output with iomanip

- The iomanip header provides tools to format how numbers and text are displayed
- setw(n): sets the minimum width of the next output field to n characters
- cout << setw(10) << 42; prints 42 right-aligned in a field of 10 characters
- setprecision(n): controls the number of digits shown for floating point numbers
- With fixed: shows exactly n decimal places
- Without fixed: shows n significant digits total
- Example: cout << fixed << setprecision(2) << 3.14159; prints 3.14
- setfill(c): fills empty spaces in a field with the character c instead of spaces
- Example: cout << setfill('0') << setw(5) << 42; prints 00042
- left and right: control alignment within the field width
- cout << left << setw(15) << "Name" prints Name followed by spaces